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-- =====  
  
-- CUSTOMER CHURN ANALYSIS  
  
-- SQL ANALYSIS QUERIES  
  
-- Project: Customer Churn Analysis - Power BI  
  
-- =====
```

```
-- =====  
  
-- 1. TOTAL CUSTOMERS  
  
-- =====
```

```
SELECT  
  
    COUNT(*) AS total_customers  
  
FROM customer_churn_1M;
```

```
-- =====  
  
-- 2. TOTAL CHURNED CUSTOMERS  
  
-- =====
```

```
SELECT  
  
    COUNT(*) AS churn_customers  
  
FROM customer_churn_1M  
  
WHERE churn = 'Yes';
```

```
-- =====  
  
-- 3. ACTIVE CUSTOMERS
```

```
-- =====
```

```
SELECT
    COUNT(*) AS active_customers
FROM customer_churn_1M
WHERE churn = 'No';
```

```
-- =====
```

```
-- 4. CHURN RATE
-- Uses a subquery
-- =====
```

```
SELECT
    churned.churn_customers,
    total.total_customers,
    ROUND(
        churned.churn_customers * 100.0 / total.total_customers,
        2
    ) AS churn_rate
FROM
    (
        SELECT COUNT(*) AS churn_customers
        FROM customer_churn_1M
        WHERE churn = 'Yes'
    ) AS churned
CROSS JOIN
    (
```

```
SELECT COUNT(*) AS total_customers
FROM customer_churn_1M
) AS total;
```

```
-- =====
-- 5. RETENTION RATE
-- Uses subqueries
-- =====
```

```
SELECT
    active.active_customers,
    total.total_customers,
    ROUND(
        active.active_customers * 100.0 / total.total_customers,
        2
    ) AS retention_rate
FROM
    (
        SELECT COUNT(*) AS active_customers
        FROM customer_churn_1M
        WHERE churn = 'No'
    ) AS active
CROSS JOIN
    (
        SELECT COUNT(*) AS total_customers
        FROM customer_churn_1M
    ) AS total;
```

```
-- =====  
-- 6. CHURN BY GENDER  
-- =====
```

```
SELECT  
    gender,  
    COUNT(*) AS total_customers,  
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) AS churn_customers,  
    ROUND(  
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)  
        * 100.0 / COUNT(*),  
        2  
    ) AS churn_rate  
FROM customer_churn_1M  
GROUP BY gender  
ORDER BY churn_rate DESC;
```

```
-- =====  
-- 7. CHURN BY CONTRACT TYPE  
-- =====
```

```
SELECT  
    contract,  
    COUNT(*) AS total_customers,  
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) AS churn_customers,
```

```

ROUND(
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*),
    2
) AS churn_rate
FROM customer_churn_1M
GROUP BY contract
ORDER BY churn_rate DESC;

```

```

-- =====
-- 8. CHURN BY PAYMENT METHOD
-- =====

```

```

SELECT
    payment_method,
    COUNT(*) AS total_customers,
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) AS churn_customers,
    ROUND(
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*),
        2
    ) AS churn_rate
FROM customer_churn_1M
GROUP BY payment_method
ORDER BY churn_rate DESC;

```

```
-- =====  
-- 9. CHURN BY TENURE GROUP  
-- =====
```

```
SELECT  
    tenure_group,  
    COUNT(*) AS total_customers,  
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) AS churn_customers,  
    ROUND(  
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)  
        * 100.0 / COUNT(*),  
        2  
    ) AS churn_rate  
FROM customer_churn_1M  
GROUP BY tenure_group  
ORDER BY churn_rate DESC;
```

```
-- =====  
-- 10. CHURN BY AGE GROUP  
-- =====
```

```
SELECT  
    age_group,  
    COUNT(*) AS total_customers,  
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) AS churn_customers,  
    ROUND(  
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
```

```

        * 100.0 / COUNT(*),

        2

    ) AS churn_rate
FROM customer_churn_1M
GROUP BY age_group
ORDER BY churn_rate DESC;

```

```

-- =====
-- 11. CHURN BY EDUCATION
-- =====

```

```

SELECT
    education,
    COUNT(*) AS total_customers,
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) AS churn_customers,
    ROUND(
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*),
        2
    ) AS churn_rate
FROM customer_churn_1M
GROUP BY education
ORDER BY churn_rate DESC;

```

```

-- =====
-- 12. CHURN BY MARITAL STATUS

```

```
-- =====
```

```
SELECT
    marital_status,
    COUNT(*) AS total_customers,
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) AS churn_customers,
    ROUND(
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*),
        2
    ) AS churn_rate
FROM customer_churn_1M
GROUP BY marital_status
ORDER BY churn_rate DESC;
```

```
-- =====
```

```
-- 13. LATE PAYMENT ANALYSIS
```

```
-- =====
```

```
SELECT
    late_payments,
    COUNT(*) AS total_customers,
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END) AS churn_customers,
    ROUND(
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*),
        2
    )
```



```
    ) AS churn_rate
FROM customer_churn_1M
GROUP BY late_payments
ORDER BY churn_rate DESC;
```

```
-- =====
```

```
-- 14. REVENUE ANALYSIS
```

```
-- =====
```

```
SELECT
    SUM(totalcharges) AS total_revenue,
    AVG(totalcharges) AS average_total_charges
FROM customer_churn_1M;
```

```
-- =====
```

```
-- 15. REVENUE LOST FROM CHURNED CUSTOMERS
```

```
-- =====
```

```
SELECT
    SUM(totalcharges) AS revenue_lost
FROM customer_churn_1M
WHERE churn = 'Yes';
```

```
-- =====
```

```
-- 16. REVENUE LOST BY CONTRACT
```

```
-- =====
```

```
SELECT
    contract,
    COUNT(*) AS churn_customers,
    SUM(totalcharges) AS revenue_lost
FROM customer_churn_1M
WHERE churn = 'Yes'
GROUP BY contract
ORDER BY revenue_lost DESC;
```

```
-- =====
```

```
-- 17. HIGH-RISK CUSTOMERS
-- Example using churn + late payments + tenure
```

```
-- =====
```

```
SELECT *
FROM customer_churn_1M
WHERE churn = 'Yes'
    AND late_payments > 0
ORDER BY totalcharges DESC;
```

```
-- =====
```

```
-- 18. HIGH-RISK CUSTOMER ANALYSIS USING SUBQUERY
```

```
-- =====
```

```

SELECT
    high_risk.total_high_risk,
    total.total_customers,
    ROUND(
        high_risk.total_high_risk * 100.0
        / total.total_customers,
        2
    ) AS high_risk_percentage
FROM
    (
        SELECT COUNT(*) AS total_high_risk
        FROM customer_churn_1M
        WHERE churn = 'Yes'
        AND late_payments > 0
    ) AS high_risk
CROSS JOIN
    (
        SELECT COUNT(*) AS total_customers
        FROM customer_churn_1M
    ) AS total;

-- =====

-- 19. TOP CUSTOMER SEGMENTS BY CHURN RATE

-- Uses CTE

-- =====

WITH churn_analysis AS

```

```
(
SELECT
    contract,
    payment_method,
    COUNT(*) AS total_customers,
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
        AS churn_customers
FROM customer_churn_1M
GROUP BY contract, payment_method
)
```

```
SELECT
    contract,
    payment_method,
    total_customers,
    churn_customers,
    ROUND(
        churn_customers * 100.0 / total_customers,
        2
    ) AS churn_rate
FROM churn_analysis
ORDER BY churn_rate DESC;
```

```
-- =====
-- 20. CUSTOMER SEGMENT WITH ABOVE-AVERAGE REVENUE
-- Uses subquery
-- =====
```

```

SELECT
    contract,
    COUNT(*) AS customers,
    AVG(totalcharges) AS average_revenue
FROM customer_churn_1M
GROUP BY contract
HAVING AVG(totalcharges) >
(
    SELECT AVG(totalcharges)
    FROM customer_churn_1M
);

```

```

-- =====
-- 21. TENURE GROUP WITH ABOVE-AVERAGE CHURN
-- Uses subquery
-- =====

```

```

SELECT
    tenure_group,
    COUNT(*) AS total_customers,
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
    AS churn_customers,
    ROUND(
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*),
        2
    ) AS churn_rate

```

```

FROM customer_churn_1M
GROUP BY tenure_group
HAVING
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*) >
(
    SELECT
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*)
    FROM customer_churn_1M
);

```

```

-- =====
-- 22. MONTHLY CUSTOMER SIGNUP TREND
-- =====

```

```

SELECT
    YEAR(signup_date) AS signup_year,
    MONTH(signup_date) AS signup_month,
    COUNT(*) AS new_customers
FROM customer_churn_1M
GROUP BY
    YEAR(signup_date),
    MONTH(signup_date)
ORDER BY
    signup_year,
    signup_month;

```

```
-- =====  
-- 23. MONTHLY CHURN TREND  
-- =====
```

```
SELECT  
    YEAR(signup_date) AS signup_year,  
    MONTH(signup_date) AS signup_month,  
    COUNT(*) AS total_customers,  
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)  
        AS churn_customers,  
    ROUND(  
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)  
        * 100.0 / COUNT(*),  
        2  
    ) AS churn_rate  
FROM customer_churn_1M  
GROUP BY  
    YEAR(signup_date),  
    MONTH(signup_date)  
ORDER BY  
    signup_year,  
    signup_month;
```

```
-- =====  
-- 24. CUSTOMER CHURN RANKING
```

-- Uses WINDOW FUNCTION

-- =====

SELECT

contract,

payment_method,

churn_customers,

churn_rate,

RANK() OVER (

ORDER BY churn_rate DESC

) AS churn_rank

FROM

(

SELECT

contract,

payment_method,

SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)

AS churn_customers,

SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)

* 100.0 / COUNT(*) AS churn_rate

FROM customer_churn_1M

GROUP BY contract, payment_method

) AS churn_summary

ORDER BY churn_rank;

-- =====

-- 25. TOP 5 HIGH-REVENUE CHURNED CUSTOMERS/RECORDS


```
-- =====
```

```
SELECT *  
FROM customer_churn_1M  
WHERE churn = 'Yes'  
ORDER BY totalcharges DESC  
LIMIT 5;
```

```
-- =====
```

```
-- 26. CHURN RATE BY GENDER USING CTE
```

```
-- =====
```

```
WITH gender_churn AS  
(  
    SELECT  
        gender,  
        COUNT(*) AS total_customers,  
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)  
        AS churn_customers  
    FROM customer_churn_1M  
    GROUP BY gender  
)  
SELECT  
    gender,  
    total_customers,  
    churn_customers,  
    ROUND(  
        (churn_customers / total_customers) * 100, 2
```

```

        churn_customers * 100.0 / total_customers,
        2
    ) AS churn_rate
FROM gender_churn
ORDER BY churn_rate DESC;

```

```

-- =====
-- 27. PAYMENT METHOD WITH HIGHEST CHURN
-- Uses subquery
-- =====

```

```

SELECT
    payment_method,
    churn_rate
FROM
(
    SELECT
        payment_method,
        SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
        * 100.0 / COUNT(*) AS churn_rate
    FROM customer_churn_1M
    GROUP BY payment_method
) AS payment_analysis
WHERE churn_rate =
(
    SELECT MAX(churn_rate)
    FROM

```

```
(
  SELECT
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*) AS churn_rate
  FROM customer_churn_1M
  GROUP BY payment_method
) AS rates
);
```

```
-- =====
-- 28. CONTRACT WITH HIGHEST REVENUE LOSS
-- Uses subquery
-- =====
```

```
SELECT
  contract,
  revenue_lost
FROM
  (
    SELECT
      contract,
      SUM(totalcharges) AS revenue_lost
    FROM customer_churn_1M
    WHERE churn = 'Yes'
    GROUP BY contract
  ) AS contract_loss
WHERE revenue_lost =
```

```
(
  SELECT MAX(revenue_lost)
  FROM
    (
      SELECT
        contract,
        SUM(totalcharges) AS revenue_lost
      FROM customer_churn_1M
      WHERE churn = 'Yes'
      GROUP BY contract
    ) AS losses
);
```

```
-- =====
-- 29. OVERALL BUSINESS KPI SUMMARY
-- =====
```

```
SELECT
  COUNT(*) AS total_customers,

  SUM(CASE
    WHEN churn = 'Yes' THEN 1
    ELSE 0
  END) AS churn_customers,

  SUM(CASE
    WHEN churn = 'No' THEN 1
```

```
ELSE 0
END) AS active_customers,
```

```
ROUND(
    SUM(CASE WHEN churn = 'Yes' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*),
    2
) AS churn_rate,
```

```
ROUND(
    SUM(CASE WHEN churn = 'No' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*),
    2
) AS retention_rate,
```

```
SUM(totalcharges) AS total_revenue,
```

```
SUM(
    CASE
        WHEN churn = 'Yes'
        THEN totalcharges
        ELSE 0
    END
) AS revenue_lost,
```

```
AVG(totalcharges) AS average_revenue
```

```
FROM customer_churn_1M;
```